WORKED SOLUTION

STEP I 2001, Question 7

Pure • Concavity and Differential Equations

STEP I 2001, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/01-s1-q7

The key idea

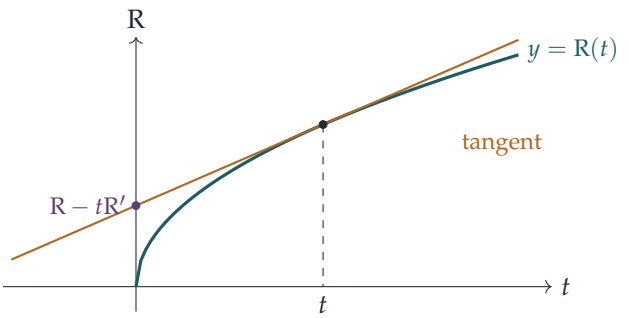
The three conditions say that R starts at the origin, rises, and bends downwards: its graph is *concave*. A concave graph lies below each of its tangents, and that single geometric fact gives the inequality in (i). Parts (ii) and (iii) are then separable differential equations, because $\mathrm{H} = \mathrm{R}'/\mathrm{R}$ is the derivative of $\ln \mathrm{R}$. Once we have R explicitly, we test it against the conditions, and the inequality from (i) gives a quick way to rule a model out.

Solution

First, a fact we use throughout: since $\mathrm{R}(0) = 0$ and R is strictly increasing for $t > 0$, we have $\mathrm{R}(t) > 0$ for $t > 0$. So $\mathrm{H}(t)$ is defined and $\mathrm{H}(t) > 0$ for $t > 0$.

Part (i)

The graph passes through the origin, has positive gradient, and the gradient decreases as t increases.



Fix $t > 0$. The tangent to the graph at time t has gradient $\mathrm{R}'(t)$, so it meets the vertical axis at height

$$\mathrm{R}(t) - t\,\mathrm{R}'(t).$$

The picture says this intercept is positive, because the curve, which passes through the origin, lies below the tangent. To make this rigorous, apply the mean value theorem to R on $[0, t]$: there is some c with $0 < c < t$ and

$$\mathrm{R}(t) - \mathrm{R}(0) = t\,\mathrm{R}'(c), \qquad \text{so} \qquad \mathrm{R}(t) = t\,\mathrm{R}'(c).$$

Since $\mathrm{R}'' < 0$, the function R' is strictly decreasing, and $c < t$ gives $\mathrm{R}'(c) > \mathrm{R}'(t)$. Hence

$$\mathrm{R}(t) > t\,\mathrm{R}'(t),$$

which is exactly the statement that the tangent's intercept is above the origin. Dividing by $\mathrm{R}'(t) > 0$,

$$t < \frac{\mathrm{R}(t)}{\mathrm{R}'(t)} = \frac{1}{\mathrm{H}(t)},$$

as required.

(Equivalently, the tangent meets the t-axis at $t - 1/\mathrm{H}(t)$, which is negative. In cosmology $1/\mathrm{H}$ is the "Hubble time", and this says that the universe is younger than its Hubble time in any decelerating model.)

Part (ii)

For $t > 0$, $\mathrm{R} > 0$, so we may write

$$\frac{\mathrm{d}}{\mathrm{d}t} \ln \mathrm{R}(t) = \frac{\mathrm{R}'(t)}{\mathrm{R}(t)} = \frac{a}{t}.$$

Integrating, $\ln \mathrm{R} = a \ln t + C$, so

$$\mathrm{R}(t) = k\,t^a \qquad (t > 0),$$

where $k = \mathrm{e}^C > 0$.

Now test the conditions, using $k > 0$ and $t^a > 0$:

- $\mathrm{R}'(t) = ka\,t^{a-1}$, which is positive for $t > 0$ if and only if $a > 0$.
- $\mathrm{R}''(t) = ka(a-1)\,t^{a-2}$. Given $a > 0$, this is negative if and only if $a < 1$.
- For $a > 0$, $kt^a \to 0$ as $t \to 0^+$, so $\mathrm{R}(0) = 0$ is consistent. (For $a \le 0$ it would fail anyway: kt^a tends to $k \neq 0$ or blows up.)

So the conditions $(*)$ hold exactly when

$$0 < a < 1.$$

This agrees with part (i): there $t < 1/\mathrm{H} = t/a$, which forces $a < 1$.

Part (iii)

Now $\mathrm{H}(t) = b/t^2$. First, $\mathrm{H}(t) > 0$ for $t > 0$ (from the opening remark), so we need $b > 0$. If $b \leq 0$ we are already done.

So suppose $b > 0$. By part (i), every $t > 0$ must satisfy

$$t < \frac{1}{\mathrm{H}(t)} = \frac{t^2}{b}, \qquad \text{that is,} \qquad t > b$$

(dividing by $t > 0$ and multiplying by $b > 0$). But this fails for every t with $0 < t \leq b$. So no value of b is consistent with $(*)$.

? **Ask yourself** Is it legitimate to use part (i) like this?

Yes. Part (i) proved that *any* function satisfying $(*)$ has $t < 1/\mathrm{H}(t)$ for all $t > 0$. So if a proposed H breaks that inequality for even one t, no function with that H can satisfy $(*)$. This is a proof by contradiction.

For a direct confirmation, solve the equation as in (ii): $\ln \mathrm{R} = -b/t + C$, so $\mathrm{R}(t) = k\,\mathrm{e}^{-b/t}$ with $k > 0$. Then

$$\mathrm{R}'(t) = \frac{kb}{t^2}\,\mathrm{e}^{-b/t},$$

$$\mathrm{R}''(t) = kb\,\mathrm{e}^{-b/t}\left(\frac{b}{t^4} - \frac{2}{t^3}\right) = \frac{kb\,\mathrm{e}^{-b/t}}{t^4}\,(b - 2t).$$

With $b > 0$, $\mathrm{R}' > 0$ and $\mathrm{R}(t) \to 0$ as $t \to 0^+$, so the first two conditions hold. But $\mathrm{R}''(t) > 0$ for $0 < t < b/2$, so the graph bends the wrong way near the origin, and $(*)$ fails.

Checks

- In (ii), $a = \frac{1}{2}$ gives $\mathrm{R} = k\sqrt{t}$, the shape in the sketch: increasing and concave, and $t < 1/\mathrm{H} = 2t$. Taking $a = 1$ gives a straight line through the origin, where $t = 1/\mathrm{H}$ exactly: the boundary case of (i).
- In (iii) with $b = 1$, $k = 1$: numerically $\mathrm{R}''(0.25) \approx 2.34 > 0$, while $1/\mathrm{H}(0.5) = 0.25 < 0.5$, both breaking $(*)$.

The computer checked (i) on several concave test functions, the solutions and the derivative formulas in (ii) and (iii) by finite differences, and the sign change of R'' at $t = b/2$.

Common slips

- In (i), arguing only from the picture. "The curve lies below its tangent" needs a reason: the mean value theorem, plus R' decreasing.

- In (ii), forgetting that $a > 0$ is needed for $\mathrm{R}' > 0$ (and for $\mathrm{R}(0) = 0$), and giving only $a < 1$.
- In (iii), claiming concavity fails for large t. The sign of $b - 2t$ shows the failure is at *small* t.

Beyond STEP

Part (i) is a special case of a central fact about convex and concave functions: a differentiable function is concave exactly when its graph lies below all its tangents. This underlies Jensen's inequality and much of optimisation. The mean value theorem argument is standard first-year analysis; see Abbott, *Understanding Analysis*. In real cosmology, $a = \frac{2}{3}$ is the matter-dominated Einstein–de Sitter universe, for which the age is two-thirds of the Hubble time.

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